CS & IT ENGINEERING

COMPUTER ORGANIZATION AND ARCHITECTURE

Instruction & Addressing Modes

Lecture No.- 06

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Recap of Previous Lecture

Topic Addressing Modes → doubt ??

Topic PC Relative Mode

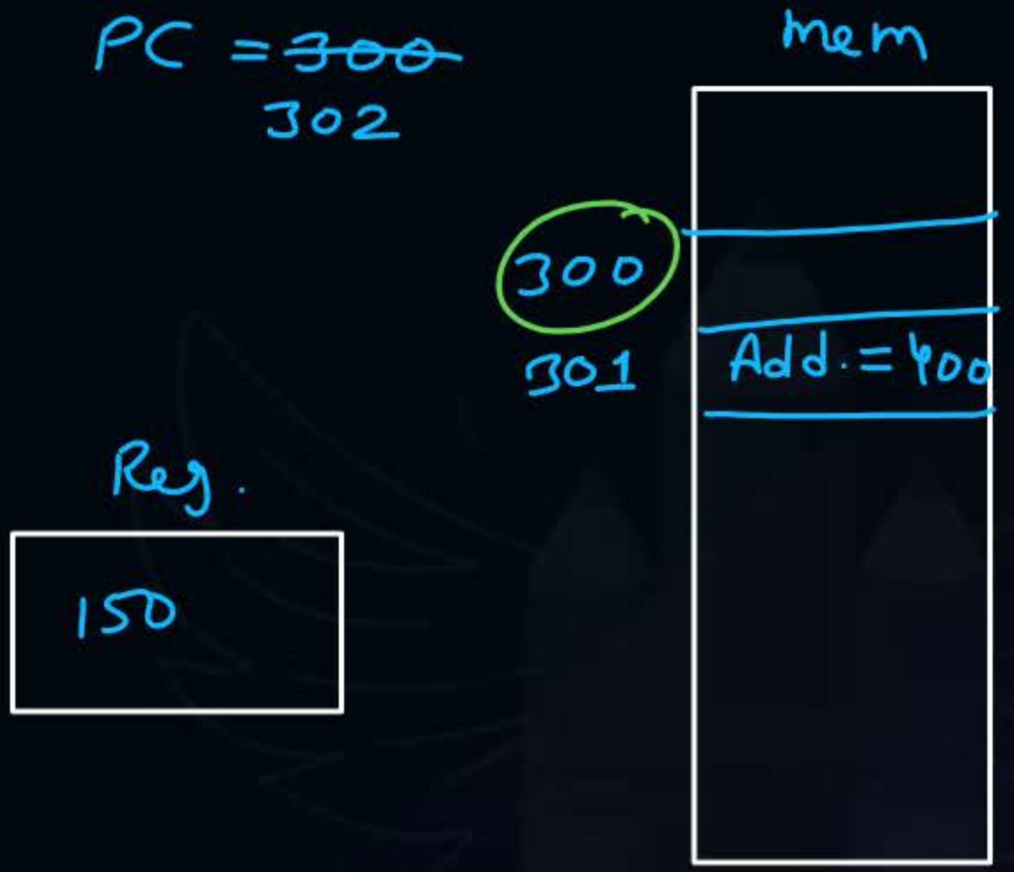
Topics to be Covered

Topic Questions on Addressing Modes

[redacted]

#Q. An instruction is stored at Location 300 with its address field at location 301. The address field has the value 400. A processor register contains the number 150. Evaluate the effective address, if addressing mode is:

1. Direct ⇒ 400
2. Immediate ⇒ 301
3. Relative ⇒ 302 + 400 = 702
4. Register Indirect ⇒ 150



[MCQ]

#Q. In case the code is position independent, the most suitable addressing mode is

A Direct mode

B Indirect mode

C Relative mode

D Indexed mode

[MCQ]

#Q. The addressing mode that permits relocation, without any change whatsoever in the code, is

A. Indirect addressing

B. Base register addressing

C. Indexed addressing

D. PC relative addressing

[redacted]

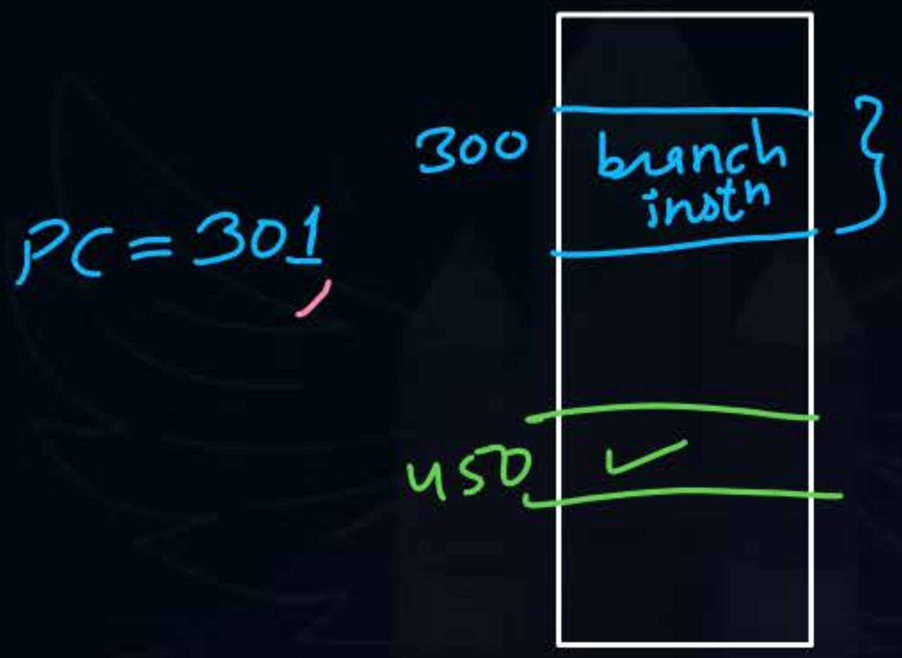
#Q. A relative branch mode type instruction is stored in memory at address 300. The branch is made to an address 450.

1. What should be the value of relative address field of the instruction? $\Rightarrow 149$
2. Determine the value of PC before instruction fetch, after the fetch and after execution phase? 300, 301, 450

Target add. $= 450 = PC + \text{offset}$

$450 = 301 + \text{offset}$

$\text{offset} = 149$



Ques) A PC-relative mode type branch instⁿ is stored in a byte addressable mem. Instⁿ length is 2 bytes and address part of instⁿ has value 160. It takes jump to an address 620. The address of the instruction in memory 458 ?

Solⁿ

Target = PC + offset

620 = PC + 160

PC = 460

add. of current instⁿ = 460 − 2

= 458 Ans.

[MCQ]

LW R1, 20(R2)

#Q. Consider a hypothetical processor with an instruction of type LW R1, 20(R2); which during execution reads a 32-bit word from memory and stores it in a 32-bit register R1. The effective address of the memory location is obtained by the addition of a constant 20 and the contents of register R2. Which of the following best reflects the addressing mode implemented by this instruction for operand in memory?

A ✗ Immediate Addressing

B ✗ Register Addressing

C ✗ Register indirect scaled addressing

D ✓ Base indexed addressing

LW R1, 20(R2)

$R1 \leftarrow mem[20 + R2]$

[NAT]

Ans = 4

#Q. Consider a three-word machine instruction

ADD A[R0], @B

The first operand (destination) "A[R0]" uses indexed addressing mode with R0 as the index register. The second operand (Source) "@B" uses indirect addressing mode. A and B are memory addresses residing at the second and the third words, respectively. The first word of the instruction specifies the opcode, the index register designation and the source and destination addressing modes. During execution of ADD instruction, the two operands are added and stored in the destination (first operand).

The number of memory cycles needed during the execution cycle of the instruction is ~~4~~___.

Destination

ADD A[R0], @B

Indirect mode

@B $\xrightarrow{\text{mem}}$ E.A $\xrightarrow{\text{mem.}}$ operand

$E.A. = A + R0 \xrightarrow{\text{mem}}$ operand

3 word $inst^n$

Opcode	Index Reg. $designat^n$	mode 1	mode 2	A	B

	operand 1	operand 2
Inst^n decode	–	–
E.A. calculat^n & operand fetch	1	2
Execution	–	–
write back	1	–
Total	4	

Ques) In prev. Quest^n if mem. is word addressable.

No. of mem. cycles in instruction cycle $\underline{\ 7\ }$?

Sol^n

$$\underbrace{3}_{\text{Inst}^n\text{ fetch cycle}} + \underbrace{4}_{\text{execution cycle}} = 7$$

[NAT]

H.W.

#Q. Consider a 6-words instruction, which is of the following type:

Opcode	Mode1	Mode2	Address1	Address2

The first operand (destination) uses register indirect mode and second operand uses indirect mode. Assume each operand is of size 2 words, each address is of 2 words and main memory takes 50ns for 1 byte access. Further assume that the opcode denotes addition operation which copies result of addition of 2 operands. One word size is 4 bytes. Total time required in:

1. Fetch cycle of instruction
2. Execution cycle of instruction
3. Instruction cycle of instruction

2 mins Summary

Topic Questions on Addressing Modes

Happy Learning

THANK - YOU